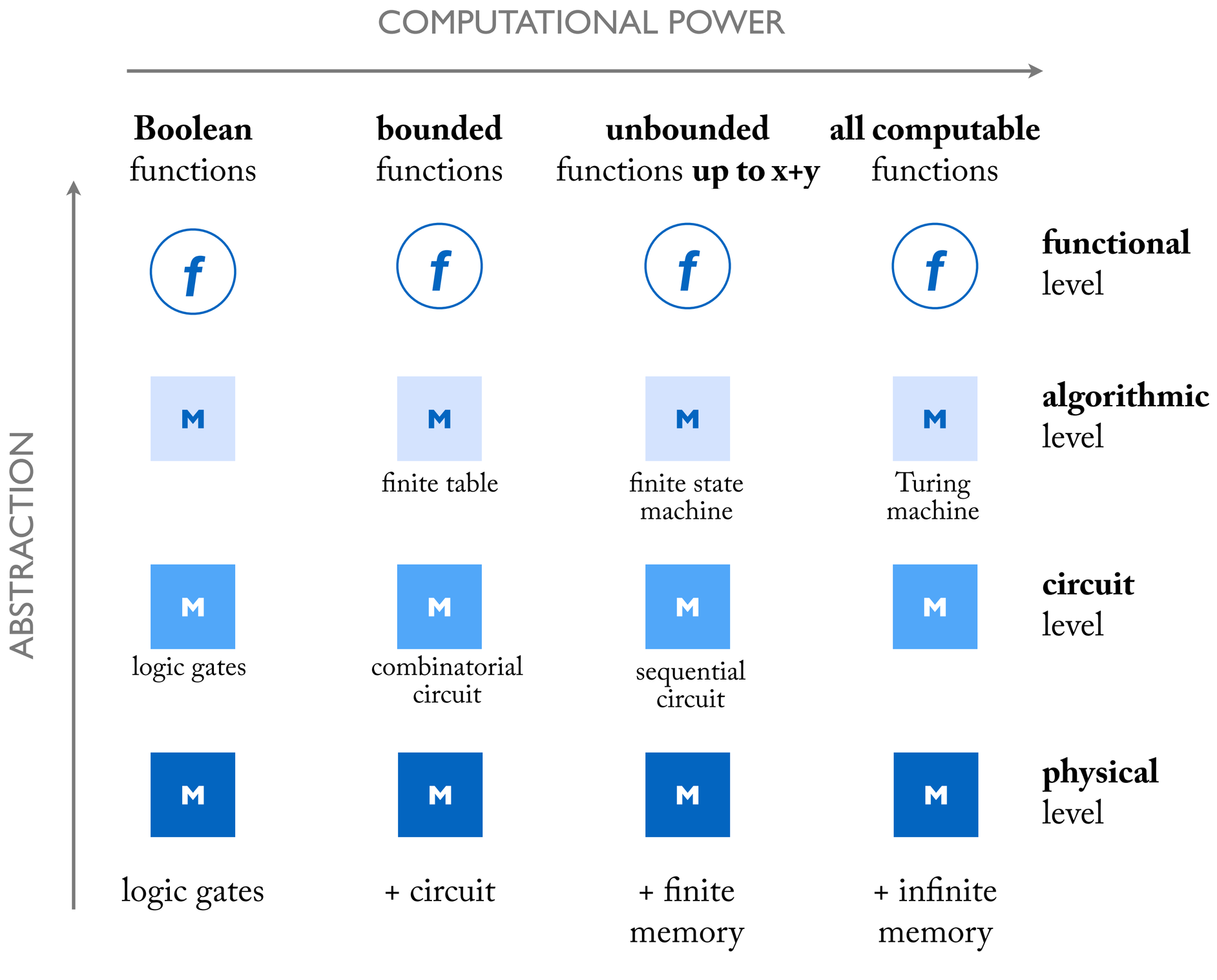
Unit 2

LEVELS

In Unit 2, we explore the dimensions of computation. We'll look at levels of abstraction and levels of computational power.



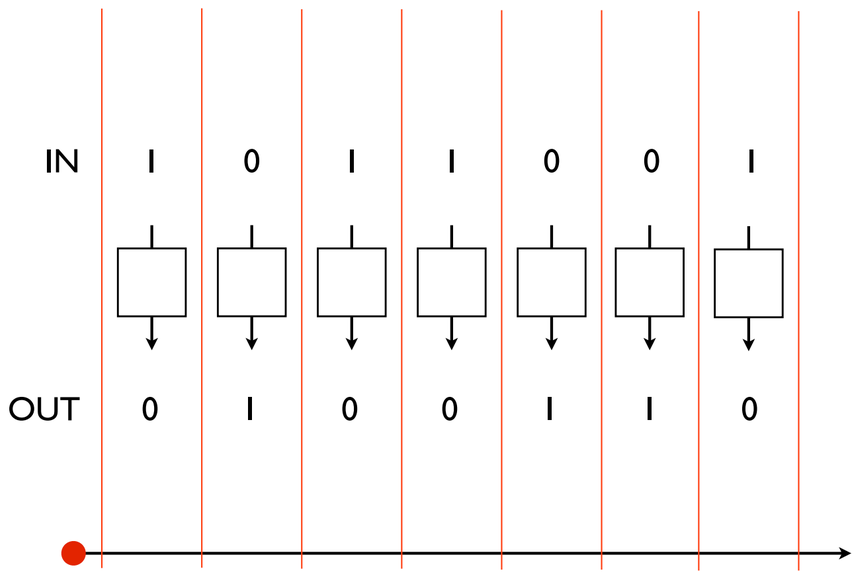
16. Memory

Harnessing the Power of Time

At the end of Unit 1, we saw some of the limitations of combinatorial circuits. For one thing, the size of the inputs they can process is limited by the size of the circuits. In a small and finite amount of space, only small and finite numerals can be processed. For another, combinatorial circuits always respond in the same way to the same input, no matter what has happened in the past. They are incapable of learning.

To overcome both problems, we must find a way to harness the power of time. If we can look back in time, we can condition our behavior on what has happened in the past. And we can access the unlimited store of time available to us for the purposes of computation. Memory, of course, is what will allow our little computers to master time.

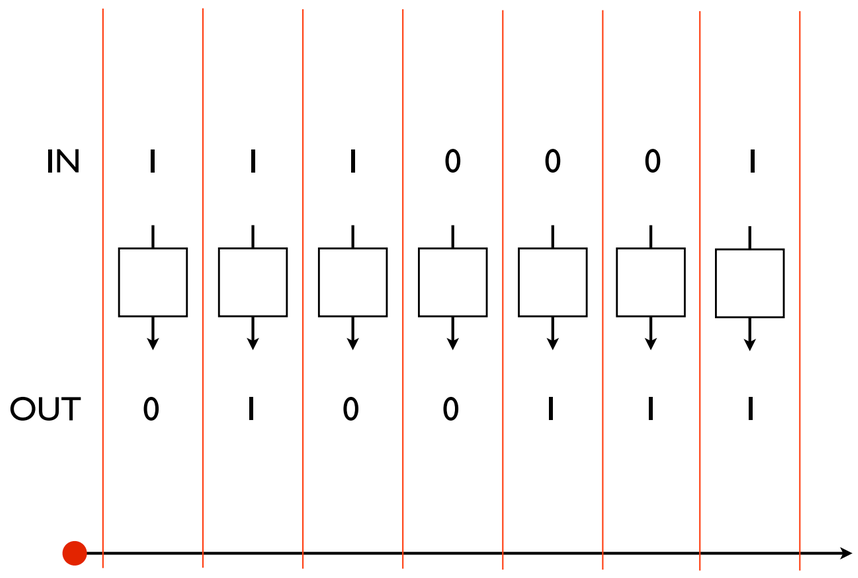
History Independent Response



This machine always responds to the same input in the same way, independent of its past responses.

No memory required.

History Dependent Response



This machine responds to the same input in different ways, depending on its past responses.

Memory required.

Memory

Memory is **information carried forward in time**. The chalkboard is a form of memory; marks are carried forward in time until they are erased. A seesaw is a primitive form of memory too; it stores information about which end was pushed down most recently.

